

Appendix 1:

Project Initiation Report

Name of the student: Arpit Kumar

Faculty mentor: Dr. Pankaj Kumar Goswami

Project start date: Feb 2026

Guide Sign:

Project Objectives:

The primary objective of the **TaxSphere – Smart Finance & Tax Intelligence System** project is to design and develop a full-stack intelligent web application that enables users to efficiently manage their personal finances and perform complete tax-related activities, including **Income Tax Return (ITR)** filing.

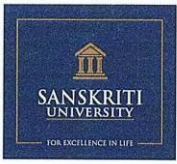
In today's digital economy, freelancers, students, and professionals often struggle with maintaining financial records, tracking expenses, and filing taxes accurately. TaxSphere aims to simplify this process by providing a unified and user-friendly platform that combines financial management, tax calculation, and ITR filing into a single system.

The application allows users to record income and expenses, set budgets, analyze financial patterns, and receive automated tax estimates. Additionally, the system includes a complete ITR filing module, where users can enter income details, apply deductions, upload required documents, and generate tax reports.

The project also integrates real-time analytics dashboards, smart financial insights, and secure payment gateway integration (Razorpay) for tax filing services. By combining automation, security, and usability, TaxSphere aims to provide a reliable and scalable solution for modern financial management.

Key Objectives:

1. Develop a modern, responsive, and user-friendly interface that works seamlessly across devices.
2. Implement secure authentication and data protection using JWT to ensure user privacy and safety.
3. Build a complete income and expense tracking system with categorization and real-time updates.



4. Design an automated tax estimation engine based on predefined tax rules and slabs.
5. Create an interactive dashboard with charts, graphs, and financial insights for better decision-making.
6. Develop a full ITR Filing Module that allows users to:
 - Enter income details
 - Apply deductions
 - Upload documents (PAN, Aadhaar, Form 16)
 - Generate ITR reports
7. Integrate payment gateway (Razorpay/Stripe) for secure tax filing and service payments.
8. Implement reporting features to export financial data in PDF and CSV formats.
9. Provide email notifications and reminders for tax deadlines and financial updates.
10. Ensure scalability so that future features like AI-based financial insights and CA integration can be added.

Project Scope and Activities:

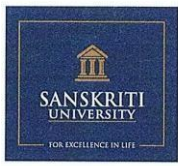
The scope of the TaxSphere project covers all major aspects of financial management and tax handling. The system acts as a digital financial assistant, combining transaction tracking, budgeting, tax estimation, and ITR filing into one platform.

The application will be developed using full-stack technologies with a modular and scalable approach. Each module will be developed, tested, and integrated step-by-step to ensure system reliability.

- **Module A: Income & Expense Management** - Recording and managing all financial transactions
- **Module B: Budget & Categorization** - Managing spending limits and categories
- **Module C: Tax Estimation Engine** - Automatic calculation of taxes based on slabs
- **Module D: ITR Filing System** - Complete process of filing tax returns
- **Module E: Payment Integration** - Secure online payment for tax filing services
- **Module F: Reporting & Analytics** - Dashboard, charts, and downloadable reports

Moreover, the project will involve research and alignment with **real-world tax policies** for accuracy. The design will prioritize **data security**, **usability**, and **speed**, ensuring that users can perform financial operations with confidence and minimal effort.

TaxSphere's scope does not end with development—it includes maintenance, deployment, and feedback-based updates to continuously refine user experience.



How the Project Will Be Performed:

Requirement Gathering:

During the initial stage, user needs and market requirements will be studied in depth. Interviews with freelancers, students, and small business owners will help identify common financial and tax-related problems. The system requirements specification (SRS) will define essential features such as income and expense tracking, budget management, tax estimation, ITR filing process, document upload, and payment integration for tax services.

UI/UX Design:

The design phase will involve wireframing the overall structure and user flow of the application. Screens for login, registration, dashboard, transaction entry, budgeting, tax module, and ITR filing will be planned in detail. The design will focus on simplicity, clarity, and responsiveness so that users can easily navigate the system across different devices and browsers.

Frontend Development:

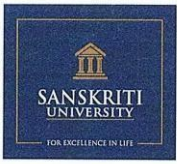
The frontend will be developed using **Angular**, ensuring a component-based structure and smooth user interaction. Features like dashboards, charts, forms, and data tables will be implemented for better visualization. Interactive elements such as financial graphs, tax charts, and ITR forms will be included to improve user understanding. The frontend will be tested for responsiveness, performance, and accessibility.

Backend Development:

Backend development will be carried out using **Node.js and Express.js**. RESTful APIs will be created to handle operations such as user authentication, transaction management, budgeting, tax calculation, and ITR processing. The system will also include **payment gateway integration (Razorpay)** to handle secure online transactions for tax filing or services. The backend will ensure scalability, reliability, and secure data handling.

Database Design:

A **NoSQL** database (MongoDB) will be used to store application data. Collections such as Users, Transactions, Budgets, TaxRecords, and ITRData will be designed. The database will ensure efficient data storage, fast retrieval, and flexibility. Security measures like validation and access control will be applied to protect user data.



Testing and Validation:

All modules will be tested using different testing methods such as unit testing, integration testing, and system testing. Sample data will be used to verify financial calculations, tax estimation accuracy, and ITR generation. The system will also be tested for performance, security, and usability.

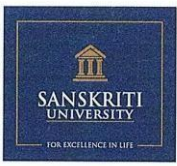
- **Deployment:** The final application will be deployed on cloud platforms. The frontend will be hosted on services like Vercel or Netlify, while the backend will be deployed on platforms like Render or AWS. MongoDB Atlas will be used for database hosting. Environment variables will be used to keep sensitive data secure.
- **Documentation and Training:** Complete documentation will be prepared including system usage, installation steps, and troubleshooting guidelines. A user guide will also be provided to help users understand how to use features like transaction entry, tax calculation, and ITR filing.

This step-by-step methodology ensures that the development is systematic, predictable, and aligned with project objectives and timelines.

Project Deliverables:

The successful completion of the **TaxSphere – Smart Finance & Tax Intelligence System** project will result in the following key deliverables:

- **Fully Functional Web Application:** A responsive and user-friendly full-stack web application that enables users, freelancers, and professionals to manage their finances, taxes, and budgets seamlessly on a single platform.
- **Secure Authentication & Database System:** Implementation of a robust authentication mechanism and a secure, encrypted database ensuring data privacy, user integrity, and protection from unauthorized access.
- **Comprehensive Functional Modules:** Fully developed modules covering income and expense tracking, budget management, tax estimation, and financial report generation.
- **Real-Time Visual Analytics Dashboard:** An interactive dashboard displaying dynamic charts and graphs to visualize income, expenses, savings, and financial trends in real time for better decision-making.
- **Tax Calculation Engine:** A configurable and scalable tax computation engine capable of automatically calculating tax based on predefined slabs, income categories, and



applicable rules.

- **ITR Filing System:** A complete Income Tax Return (ITR) filing module that allows users to:
 - Enter income details
 - Apply deductions (80C, 80D, etc.)
 - Upload documents (PAN, Aadhaar, Form 16)
 - Generate structured ITR reports in downloadable format
- **Payment Integration System:** Integration of a secure payment gateway (Razorpay/Stripe) to handle online transactions for tax filing services, CA consultation, or other related services.
- **Automated Report Generation System:** Functionality to generate and export financial summaries and tax reports in PDF and CSV formats for record-keeping and official use.
- **Email & Alert Notification System:** Integrated email and alert mechanism to notify users of quarterly tax deadlines, upcoming payments, and budget overspending alerts.
- **Version-Controlled Source Code Repository:** Well-documented and organized source code maintained on GitHub with proper version control and development logs.
- **Cloud-Based Deployment Package:** A fully deployed web application hosted on a secure, cloud-based platform ensuring scalability, performance optimization, and accessibility.
- **Comprehensive User Documentation:** A detailed user guide/manual with screenshots and explanations for each feature, assisting end-users in easy navigation and usage of the system.

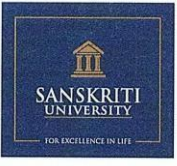
Conclusion

TaxSphere is a modern and smart system designed to make financial management and tax-related tasks simple and efficient. It provides a single platform where users can track their income and expenses, analyze their financial data, and calculate taxes easily.

The system also includes a complete ITR filing module, which allows users to enter their financial details, upload required documents, and generate tax reports without using multiple platforms.

With features like a real-time dashboard, automated tax calculation, and secure payment integration, TaxSphere helps users manage their finances in a better and more organized way. The use of modern technologies ensures that the system is fast, secure, and user-friendly.

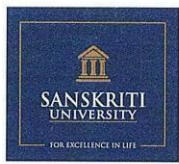
Overall, the project successfully achieves its goal of simplifying financial management and tax



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filing. In the future, it can be further improved by adding advanced features like AI-based financial suggestions, mobile application support, and integration with government tax systems.



Mid-term checkpoint review

Name of the student: Arpit Kumar

Faculty mentor: Dr. Pankaj Kumar Goswami

Project start date: 2 – Feb - 2026

Checkpoint date: 17- Feb-2026

Guide Sign:

Activities completed:

During the mid-term phase of the TaxSphere project, significant progress has been achieved in both frontend and backend development. The project objectives, system architecture, and module design have been finalized successfully. The frontend has been developed using Angular with a responsive and user-friendly interface, while the backend has been configured using Node.js and Express.js to handle API operations efficiently. The MongoDB database has been designed and implemented to store user data, transactions, and tax-related records securely.

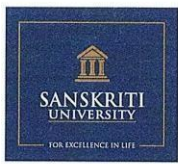
A secure authentication system using JWT has been developed and tested successfully for login and registration. Core modules such as income and expense tracking, budget management, and the financial dashboard have been implemented. The dashboard includes charts and visual analytics for better understanding of financial data.

The Tax Estimation Module has been partially implemented for automatic calculation of taxes based on user inputs. The **ITR Filing Module** has also been initiated, including features like income entry, deduction handling, and document upload. Additionally, **payment gateway integration (Razorpay)** has been set up for handling secure transactions. Email integration using Nodemailer has been configured for sending notifications and alerts. The project source code has been maintained on GitHub for version control and collaboration.

Implemented modules for:

The TaxSphere project has successfully implemented several important modules that form the core functionality of the system. The Authentication Module has been developed to provide secure user registration and login using JWT-based authentication. The Transaction Module allows users to add, edit, and delete income and expense records, helping them manage their financial data efficiently.

The Dashboard Module has been implemented to display real-time financial summaries such as total



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income, expenses, savings, and cash flow, along with visual analytics using charts. The Budget Module helps users plan and control their spending by setting limits and tracking expenses.

The Tax Estimation Module has been partially implemented to automatically calculate taxes based on user inputs and predefined tax rules. The ITR Filing Module has also been developed with core features such as income detail entry, deduction handling, and document upload functionality.

In addition, the system includes Payment Integration (Razorpay) for secure online transactions related to tax filing or services. The Notification Module has been implemented using email services to send alerts and reminders. All modules are connected with a MongoDB database and are managed through a structured backend using Node.js and Express.js.

Activities Stalled or Delayed:

Some parts of the TaxSphere project are still under development and have been slightly delayed. The **Tax Estimation Module** needs further improvement to handle different tax rules and ensure accurate calculations. The **ITR Filing Module** is partially completed, but features like final report generation and proper validation are still in progress.

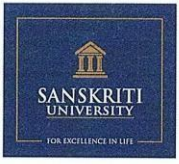
The **payment integration (Razorpay)** has been set up, but full testing in real transaction scenarios is pending. The **report generation feature** (PDF and CSV) is also not fully completed and requires additional work. The **email notification system** needs further testing to ensure that alerts and reminders are delivered correctly.

Additionally, the **final deployment of the project** has been delayed until all modules are fully tested and stable. These delays are mainly due to the complexity of integrating multiple features and ensuring system accuracy and security.

Suggested Interventions:

To complete the TaxSphere project successfully, the focus should be on finishing the remaining modules and improving system accuracy. The **Tax Estimation Module** should be refined to ensure correct calculations based on different tax rules. The **ITR Filing Module** needs to be completed by adding proper validation and final report generation features.

The **payment integration (Razorpay)** should be tested properly in both test and live environments to ensure secure and smooth transactions. The **report generation feature** (PDF and CSV) should be completed and verified for correct data output. The **email notification system** should also be tested to make sure users receive alerts and reminders on time.



Appendix 2:

In addition, proper testing of all modules should be carried out to remove errors and improve performance. Finally, the system should be deployed on a cloud platform after successful testing, and user documentation should be prepared for easy understanding and usage.

Summary:

The TaxSphere project is progressing well, with most of the core features already developed and working properly. Important modules such as authentication, income and expense tracking, dashboard, and basic tax estimation have been successfully implemented. The system is able to manage financial data and provide useful insights to users through a simple and user-friendly interface.

Work has also started on advanced features like the ITR filing module and payment integration, which are in the final stages of development. Some features such as report generation, email notifications, and final testing are still in progress and will be completed in the next phase.

Overall, the project is on the right track, and with proper testing and final improvements, it will be ready for deployment as a complete financial and tax management system.

